

SQL & Excel Assignment - Questions and Answers

Q1. Create table SalesData and insert 10,000 rows.

```
CREATE TABLE SalesData (  
  CustomerID INT,  
  Name VARCHAR(100),  
  Age INT,  
  City VARCHAR(50),  
  PurchaseAmount DECIMAL(10,2),  
  PurchaseDate DATE  
);  
  
INSERT INTO SalesData  
SELECT  
  ROW_NUMBER() OVER (ORDER BY (SELECT NULL)),  
  CONCAT('Customer_', ABS(CHECKSUM(NEWID())) % 10000),  
  ABS(CHECKSUM(NEWID())) % 50 + 18,  
  CHOOSE(ABS(CHECKSUM(NEWID())) % 5 + 1,  
    'Delhi', 'Mumbai', 'Pune', 'Chennai', 'Bangalore'),  
  RAND(CHECKSUM(NEWID())) * 10000,  
  DATEADD(DAY, -ABS(CHECKSUM(NEWID())) % 365, GETDATE())  
FROM sys.objects a  
CROSS JOIN sys.objects b;
```

Q2. Find total sales per city and top 5 cities by revenue.

```
SELECT City, SUM(PurchaseAmount) AS TotalSales  
FROM SalesData  
GROUP BY City;  
  
SELECT TOP 5 City, SUM(PurchaseAmount) AS Revenue  
FROM SalesData  
GROUP BY City  
ORDER BY Revenue DESC;
```

Q3. Find customers with purchases above average.

```
SELECT *  
FROM SalesData  
WHERE PurchaseAmount >  
(  
  SELECT AVG(PurchaseAmount)  
  FROM SalesData  
);
```

Q4. Identify and handle missing values.

Missing values found in:
Customer_Name, Region, Product, Sales, Quantity.

Handle missing values:

- Numeric columns -> Fill with Average
- Text columns -> Fill with 'Unknown'

Q5. Identify and remove duplicates.

51 duplicate rows found.

Excel Steps:
Data -> Remove Duplicates

Q6. Generate chart and identify highest sales region.

Highest Sales Region:
West Region

Interpretation:
West region has the highest sales performance.

Q7. Create pivot table showing total sales by region and product.

Pivot Table Summary:

West region recorded highest overall sales.
Monitor product generated highest revenue.

Q8. Generate calculated column Total Sales.

Formula:
=Sales * Quantity

Corrected Formula:
=VALUE(E2)*D2

Q9. Summarize dataset and key insights.

Key Insights:

- Highest sales region: West
- Best selling product: Monitor
- 51 duplicate rows found
- Missing values present
- Quantity column contains invalid text values

Q10. Generate VBA code for cleaning data.

```
Sub CleanData()  
ActiveSheet.UsedRange.RemoveDuplicates Columns:=Array(1,2,3,4,5,6), Header:=xlYes  
Cells.SpecialCells(xlCellTypeBlanks).Value = "Unknown"  
Cells.Columns.AutoFit  
Rows(1).Font.Bold = True  
End Sub
```